Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Course Information

Prof Lee Wee Sun

Objectives

The course introduces students to the analysis and design of algorithms.

Objectives

Upon completion of the course, students should be able to:

- Perform analysis of the asymptotic performance of algorithms.
- Design efficient algorithms to solve problems.
- Demonstrate knowledge of common algorithms.

Syllabus

- Asymptotic analysis
- Divide and conquer
- Randomized algorithms and quicksort
- Lower bound on sorting
- Linear time sorting
- Select
- Amortized analysis

Syllabus

- Dynamic programming
- Greedy algorithms
- NP-completeness
- Approximation algorithms

Course Staff

Lecturer: Prof Lee Wee Sun

leews@comp.nus.edu.sg

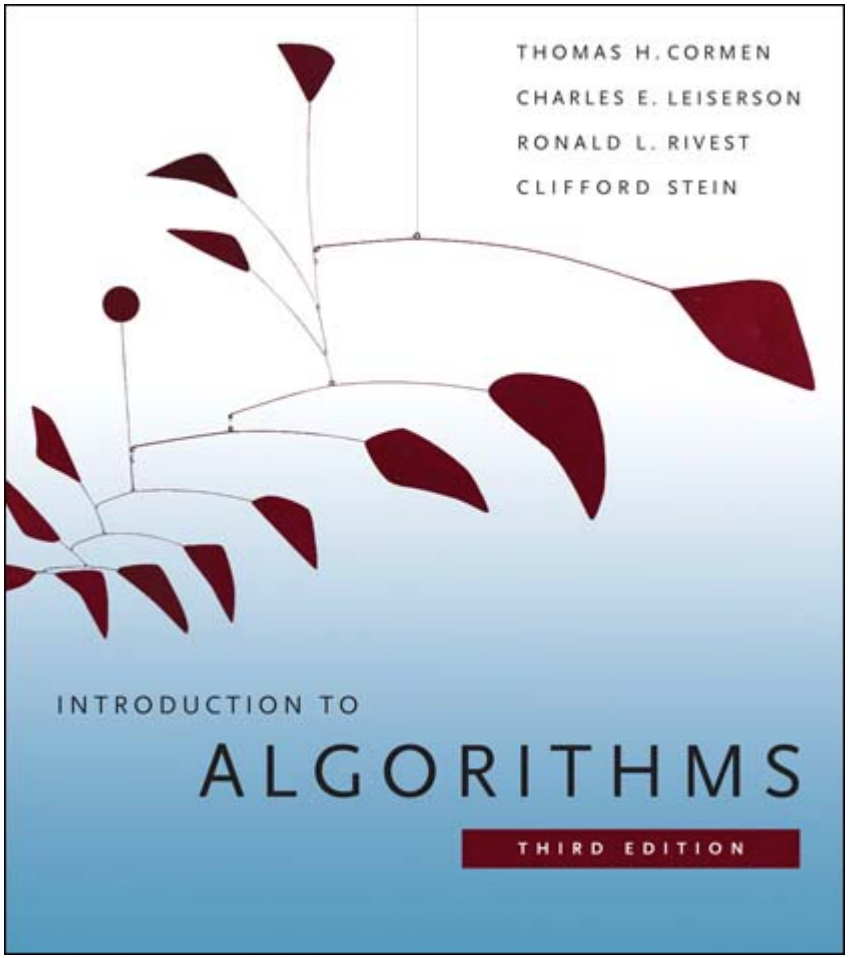
Teaching assistants:

- Erick Purwanto (Fri 5pm, 7pm)
  a0050717@u.nus.edu.sg
- Oh Shunhao (Thurs 4pm, Fri 3pm)
  a0065475@u.nus.edu.sg
- Li Juekun (Thurs 5pm, Fri 4pm)
  a0109705@u.nus.edu
- Hou Ruomu (Wed 12noon, 1pm)
  a0131421@u.nus.edu

Textbook

Introduction to Algorithms

Cormen, Leiserson, Rivest, Stein (CLRS)



Mode of Learning

Flipped classroom: Watch video before coming to class (the week before)

- Recitation (Optional):
  - Tuesdays 10am-12noon
  - Work through MCQs in class (bring your computer or mobile device)
    - Graded for effort (awarded for completion)
    - If not attending, complete on your own before end of week
  - Work through structured questions in class (not graded)
    - Work through yourself if not attending, solution provided

Mode of Learning

- Tutorial (Graded):
  - Work through MCQs in class (bring your computer or mobile device)
    1. Work individually for a few minutes. Input answer.
    2. TA shows answer distribution
    3. Discuss solution with your group
    4. Lock in your final answer.
    5. TA shows answer distribution and facilitate whole class discussion.

Mode of Learning

Tutorial grades: For each question

- Answer individually after discussion
- Grading 75% individual 25% group
- **Cooperate** to get correct solution for all group members
- Disagreement results in lower group grade (unless the group agrees on wrong answer ☺)

Mode of Learning

Flipped Classroom: Use technology to optimize learning

Not much interaction possible during large class lecture, so use videos instead

- Can rewind to view difficult concepts again
- Can play at faster speed if you are able to absorb the material faster

Mode of Learning

Flipped Classroom: Use technology to optimize learning

Use contact time for problem solving, interaction with peers, interaction with instructor

- Recitation: Problem solving, some peer interaction, some instructor interaction
- Tutorial: Small class. Problem solving, more peer and instructor interaction.

Mode of Learning

Homework and Programming Assignments

- Encouraged to work in groups
  - But try to solve yourself (e.g. spend at least 30 mins per question solving the problem) before having group discussion
  - Discuss only ideas/insights

Mode of Learning

Homework and Programming Assignments

- Encouraged to work in groups
  - **Must** work out details and write up homework or program yourself – **plagiarism** not tolerated.
  - Please do not put your code in public repositories

Mode of Learning

Homework and Programming Assignments

- Your are also encouraged to discuss ideas/insights in the online forum.
  - Similarly, try the problem yourself before discussion, and write up yourself.

Grading Scheme

Continuous Assessment (40%)

Midterm test (20%)

- Saturday of Week 7

Final exam (40%)

- 22 November 2016 evening

Continuous Assessment

Mostly formative (encourage learning) rather than summative (evaluation)

- Use Coursemology, a gamified platform
- Earn EXP for learning activities (class activities, homeworks, programming assignments)
- Many activities graded for completion (effort)
- More EXP available than required, so should focus on learning even when graded for correctness.

Continuous Assessment

Activity	Number	EXP	Total Activity EXP
Pre-requisite Test	1	1000	1000
Homework	4	3000	12000
Tutorial	10	1000	10000
Recitation	12	1000	12000
Programming Assignment	3	3000	9000
Midterm Revision	1	1000	1000
Final Revision	1	1000	1000
Total EXP			46000

Continuous Assessment

- Each 1000 EXP earns 1 mark.
- 46000 EXP available. Some additional EXP available through
  - Some side quests (harder exercises)
  - R-module exercises
  - Forum participation
    - Top forum participant each month (posts + likes) gets 100 EXP * number of weeks, second top gets 90 EXP … , …, 10th most active get 10 EXP * num of weeks.
- CA mark capped at 40, even if more than 40000 EXP earned

R-module

- CS3230R: 1MC module run in parallel with CS3230
- More depth on essentially the same topics
  - Divide and conquer in different areas
  - Randomized algorithms in data structures
  - Amortized analysis in online algorithms
  - Complexity assumptions in cryptography

R-module

To qualify to do R-module:

- Watch the corresponding video lectures (at least 5 out of 7) and complete the exercises with score of at least 80% each
- Do at your own pace, best after the corresponding topics in lecture as indicated in Lesson Plan
- Exercises give a small amount of additional EXP, even if you are not registering for the R-module

R-module

R-module grading:

- Entirely on Take Home Exam in Week 12
- Must complete at least 4 out the required 5 exercises by start of Week 12 to qualify for the take home exam (and complete all the required exercises by end of reading week)

Those who complete CS3230R will have the course registered next semester (instead of this one).

Have Fun!